

AI-Python Final Project

Updated: Week 8, Spring 2026

Instructor: Yan-Bin Chen (陳彥賓)

Master Program in Statistics, Center for General Education

National Taiwan University, Taipei, Taiwan

- **Group-based presentation:**

The final project is organized and presented in groups. Group members are formed by the students themselves.

-- groups are organized by **3 to 7 people**, with a maximum of **14 groups**.

- **Important date:**

Week 8, announce final project. Students start to form the groups.

Week 10, complete and report your grouping decision to TAs

Week 14, presentation I (8 groups)

Week 15, presentation II (8 groups)

Week 16, submit all materials to NTU COOL.

- **Grouping decision:**

Please report following information to TAs

-- Group members and student ID.

-- Presentation date.

-- Which path will you take as your final project? Path 1 is the provided data and path 2 is your own data.

- **Presentation:**

The presentation should be about 10 minutes. Path 2 groups may take up to 20 minutes. During the presentation, we will ask a few questions about your work.

- **Workload distribution:**

Please add one to two slides explaining how **the workload was distributed** among your team members.

For instance:

John T11100001	Wrote the main part of the code; delivered part of the oral presentation.
Alex T11100002	Wrote the minor parts of the code; integrated and consolidated the work; delivered part of the oral presentation.
Mary T11100003	Created the slides; analyzed the mathematical forms; delivered part of the oral presentation.
June T11100004	Surveyed and collected materials; delivered part of the oral presentation.
Chen T11100005	Just came to join.

- **Material submission:**

Please submit two files: (1) [presentation slides](#) and (2) [python codes](#) to NTU COOL.

[Submit by one member to represent your group, instead of by each member.](#)

[Please name your files using the following format with your Group number:](#)

[Group01_slides_XXXX.XXX](#) (“XXXX.XXX” means whatever name you prefer)

[Group01_codes_XXXX.XXX](#)

- **Two paths:**

Path 1: Provided dataset. We provide two datasets. Your goal is to accomplish the tasks we assigned in the following table. The Path 1 is proper to the ones who are the beginners.

Path 2: Your own dataset. If you have your own dataset, you may showcase your accomplishments of it. For Path 2, you may introduce your datasets in the beginning, then specify the goal you hope to achieve. For instance, you may say your goal is to predict something, visualization of something, data cleaning, or conduct the statistical analysis on something, etc. You could discuss with me about the main goal you will achieve.

Path 1: Simple datasets are provided.

[Please make sure your slides cover all contents from Task 1 to Task 4. However, you will only present Task 3 and Task 4. There is no need to present Task 1 and Task 2.](#)

Task 1. Taiwan Air Quality Observation.

The file `'air_quality.csv'` contains the Taiwan Air Quality Observation Data, which records hourly air quality measurements across Taiwan from 2018 to 2024, with a total of 5,882,208 entries. The dataset includes the following columns: *SiteName*: Monitoring station name, *County*: County or city name, *PublishTime*: Observation timestamp (hourly) and *PM2.5*: Fine particulate matter smaller than 2.5 μm ($\mu\text{g}/\text{m}^3$). (More details of columns: [Taiwan Air Quality Observation Data \(2018–2024\)](#)).

(T1-a) We know that when the PM2.5 concentration exceeds 35 $\mu\text{g}/\text{m}^3$, the air quality becomes unhealthy for sensitive groups such as the elderly, children, and individuals with heart or lung diseases.

Please determine: During August 2024, in Taipei City, how many days were considered unsafe for these sensitive groups to go outdoors? (Hint: You may need to use *loc* or *iloc* to filter rows that meet conditions.)

(T1-b) Visualize the monthly air pollution levels (e.g., average PM2.5) for all counties in Taiwan. Based on your visualization, identify: Which city or county had the worst air quality. Which city or county had the cleanest air?

Task 2. Image Restoration

You are asked to complete the following exercises using the file `'corrupted_images.zip'`. The file contains 10 color images in CSV format, compressed into a single archive. Each CSV file represents one image, where each record (row) contains three color channel values: Red (R), Green (G), and Blue (B), corresponding to a specific coordinate (x, y).

(T2-a) List any incomplete or invalid pixel data that contain missing values or values outside the valid range of 0 to 255. (Hint: You may use `np.where()` or pandas logical filtering to identify unreasonable pixel values (e.g., $R < 0$, $R > 255$)).

(T2-b) Based on your findings in (T2-a), handle all missing or invalid pixel data using appropriate strategies. (Hint: You may replace out-of-range values by using the average of their neighboring valid pixels, or apply other reasonable data-cleaning techniques.)

Task 3. **TBD**

Task 4. **TBD**

Path 2: Analyze your own datasets.

Your final project should describe the state of the data before and after your data processing efforts. The accomplishment of handling this data is the focus of your final project. For example, you might begin by explaining that you acquired raw data without any preprocessing. Over the following days, you implemented various techniques to reduce noise and improve data quality. In your final presentation, you'll showcase the cleaned dataset. This progression will serve as the central theme of your final project, titled *'Cleaning Noisy Data in XXX.'*